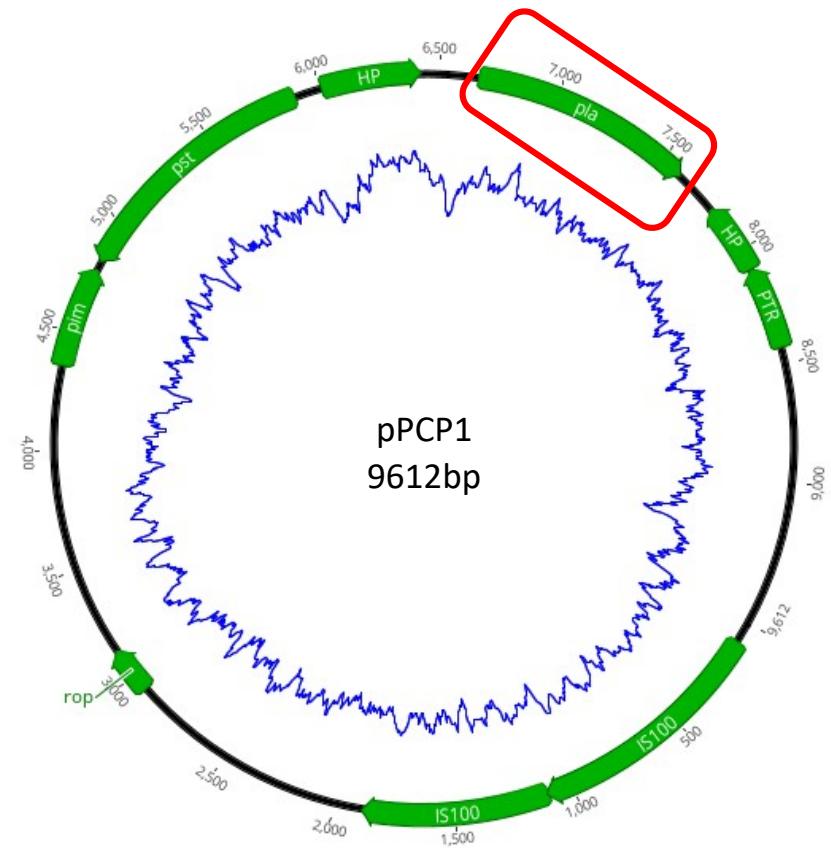
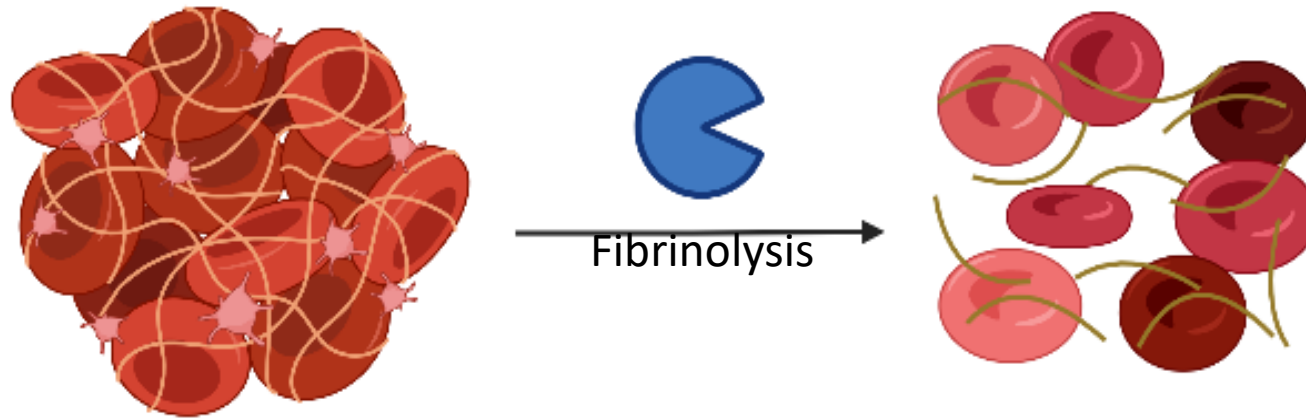


pPCP1 and *pla*

- High copy number plasmid
- Encodes PLA virulence factor
- Required for the development of bubonic & pneumonic plague



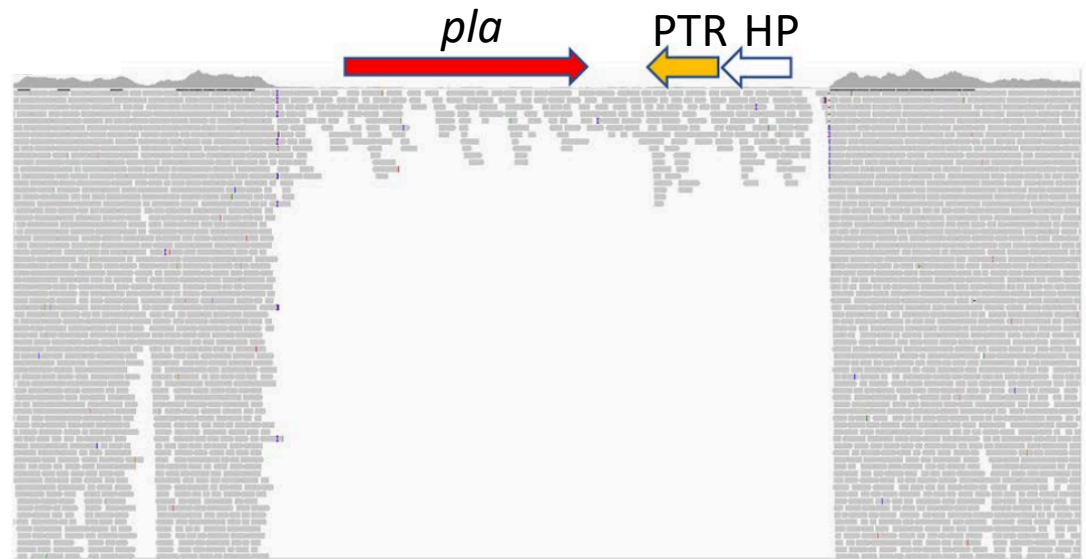
Plasminogen activator (Pla)



Blot clot preventing
Y. pestis dissemination

pla depletion

- Susat et al., (2020) – noted *pla* depletion, based on decreased % coverage within the *pla* region (~2000bp)

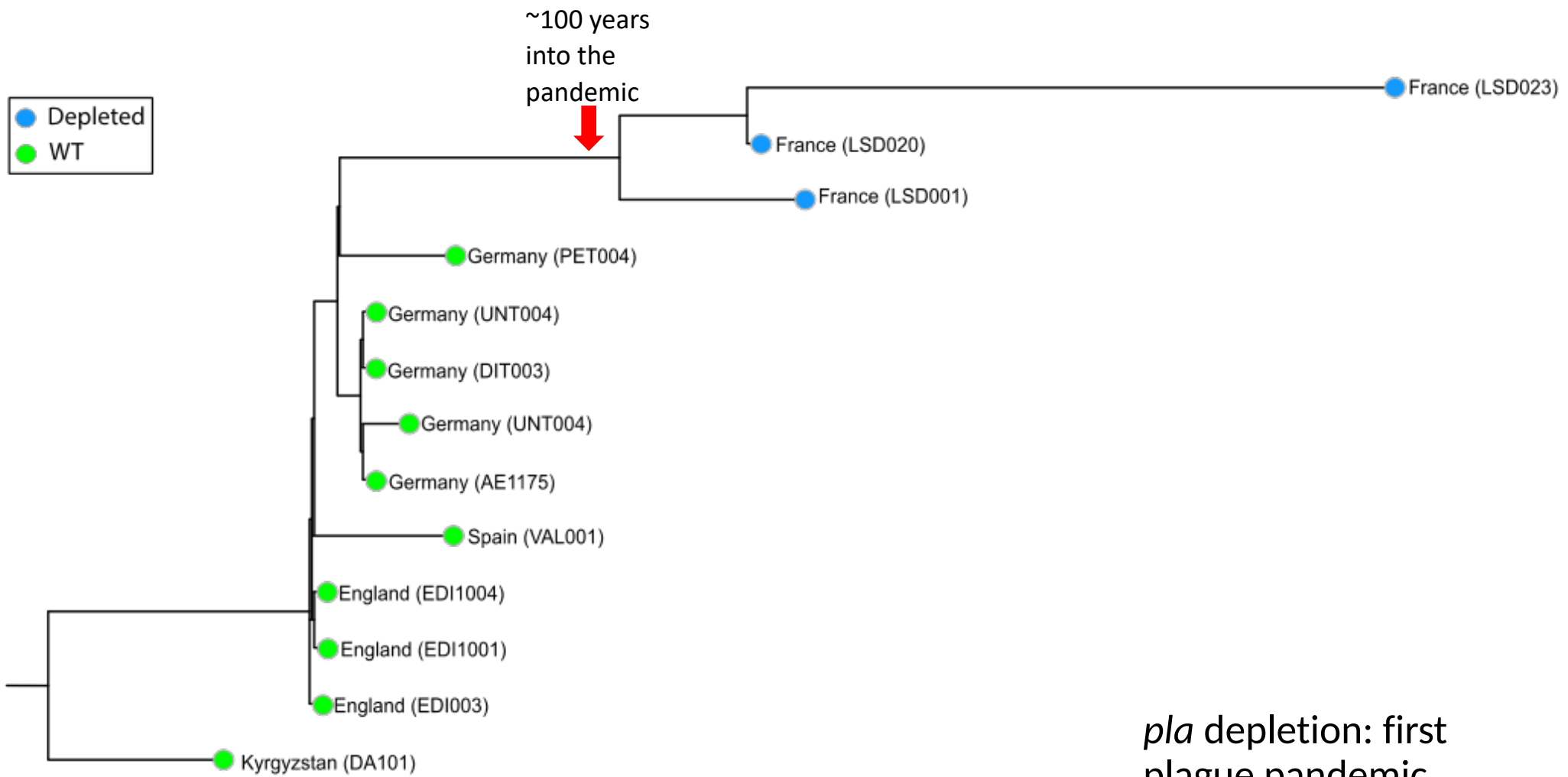


Susat et al.,
(2020)



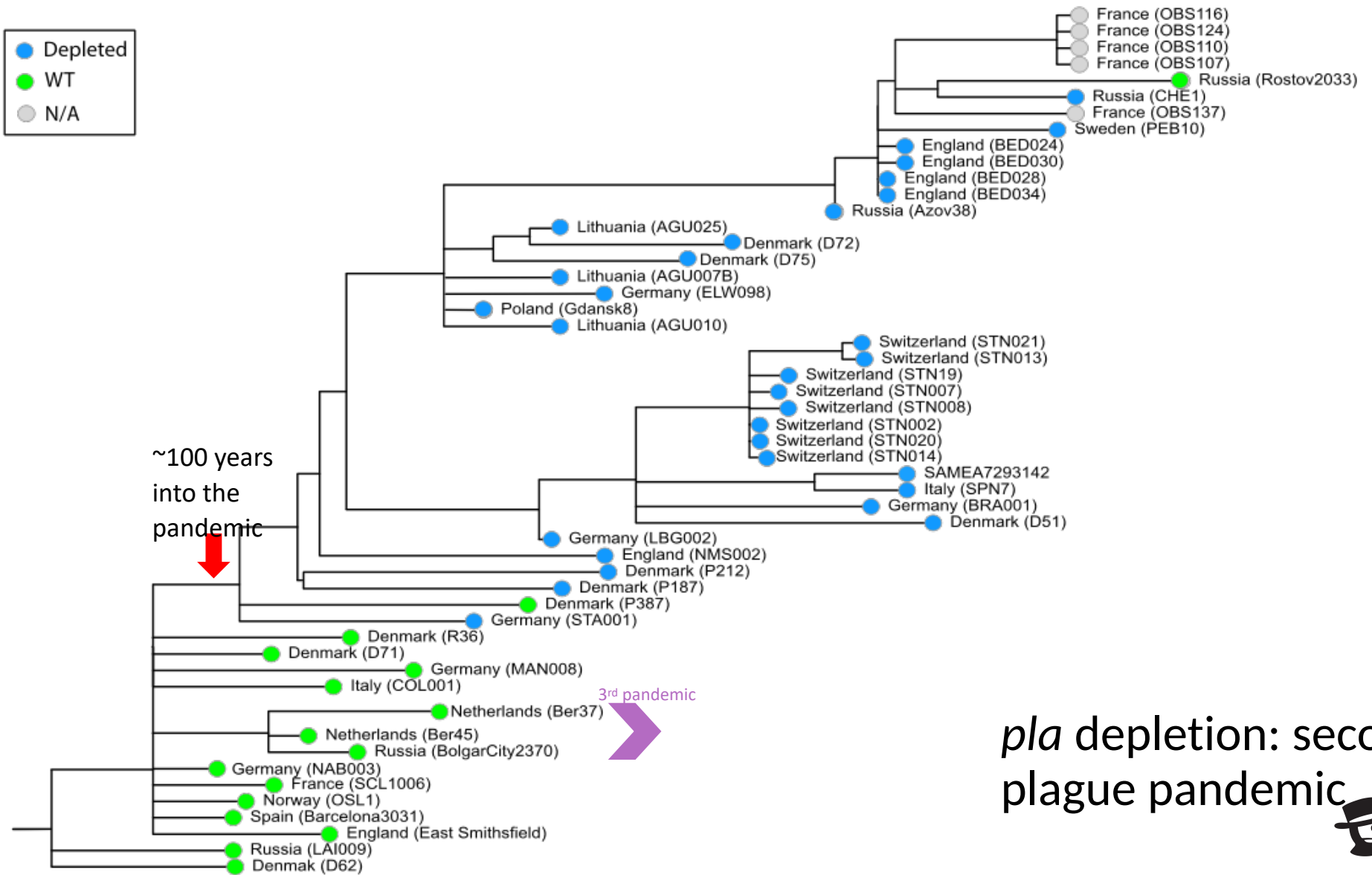
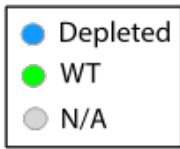
What the data show:

1. There is a convergent depletion of the pla gene in the first, second and to a very limited and localised extent in the third pandemic - data are very very clear.



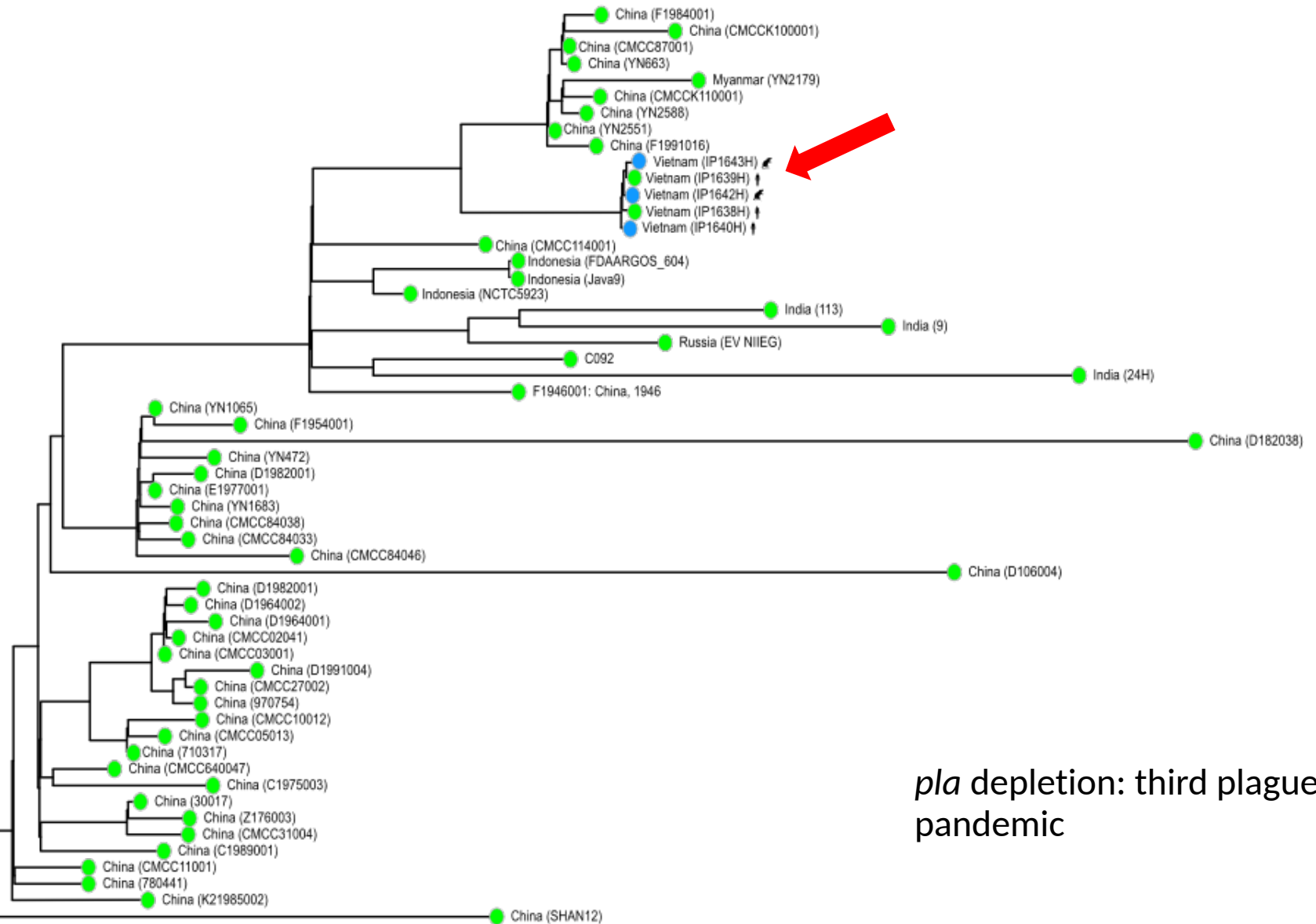
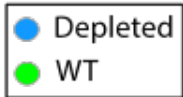
pla depletion: first
plague pandemic





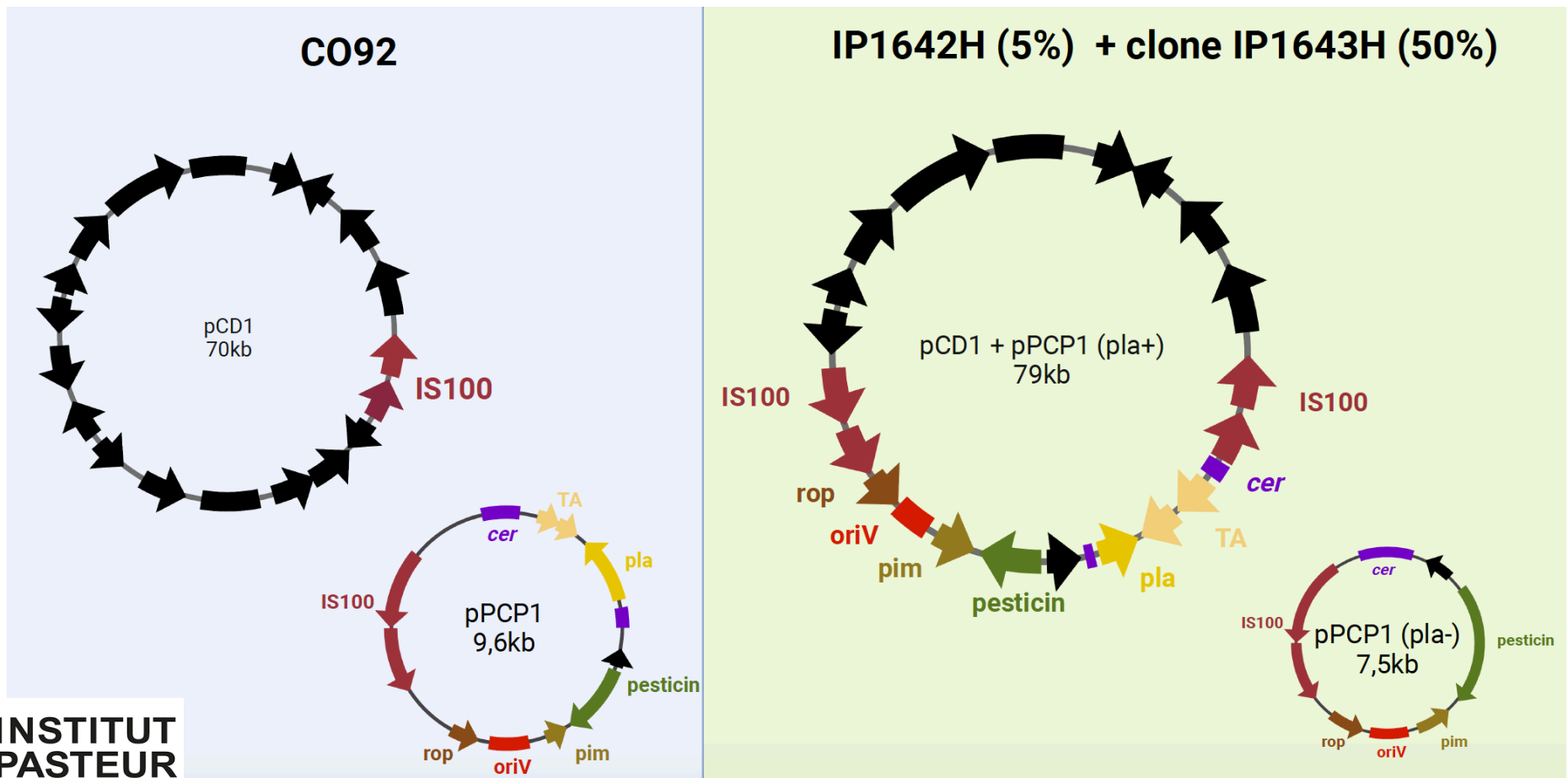
pla depletion: second plague pandemic





pla depletion: third plague pandemic

Minlon sequencing (long reads)



What the data show:

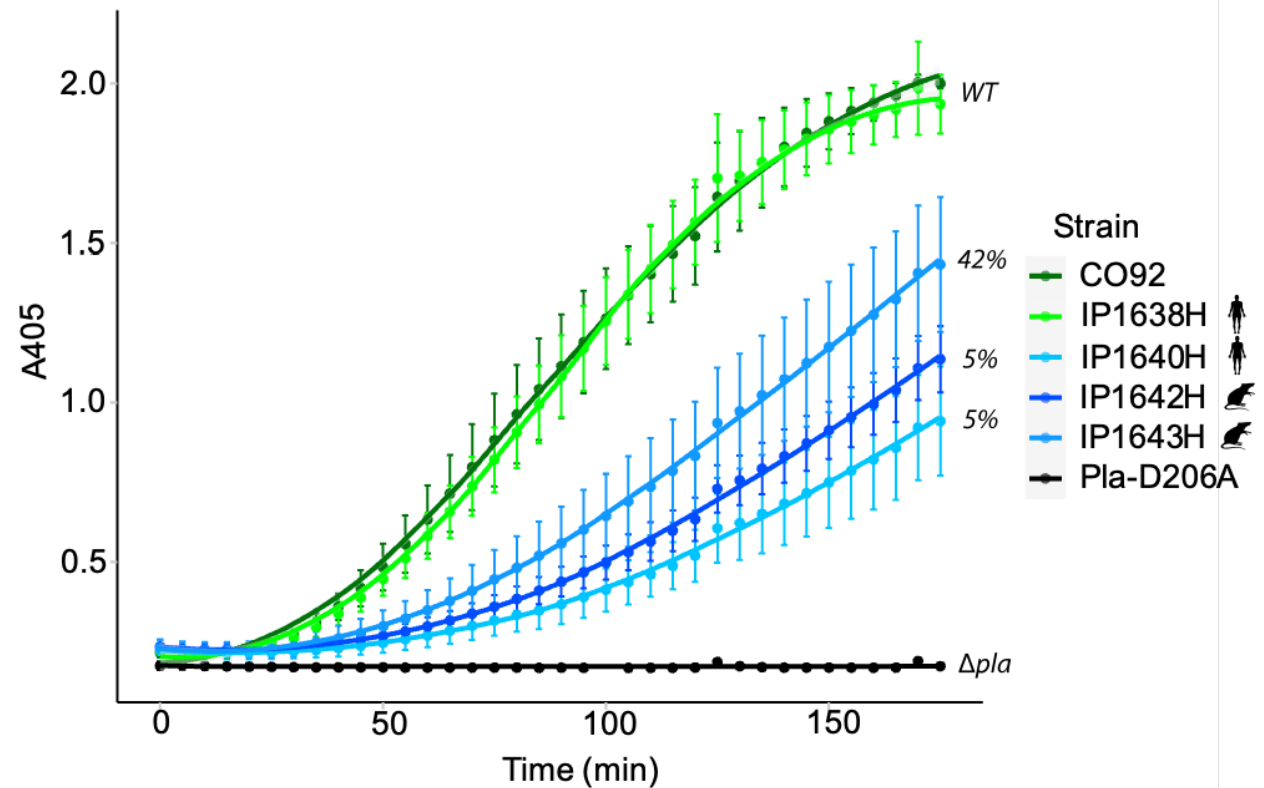
3. Enzymology with pla depleted strains show's reduced activity - matches the amount of depletion

What the data show:

1. There is a convergent depletion of the pla gene in the first, second and to a very limited and localised extent in the third pandemic - data are very very clear.
2. Very unlikely to be coincidence

Plasminogen activation assay

- Decreased enzymatic activity of Pla in depleted strains
- Activity levels correlate directly with genomic content of *pla*

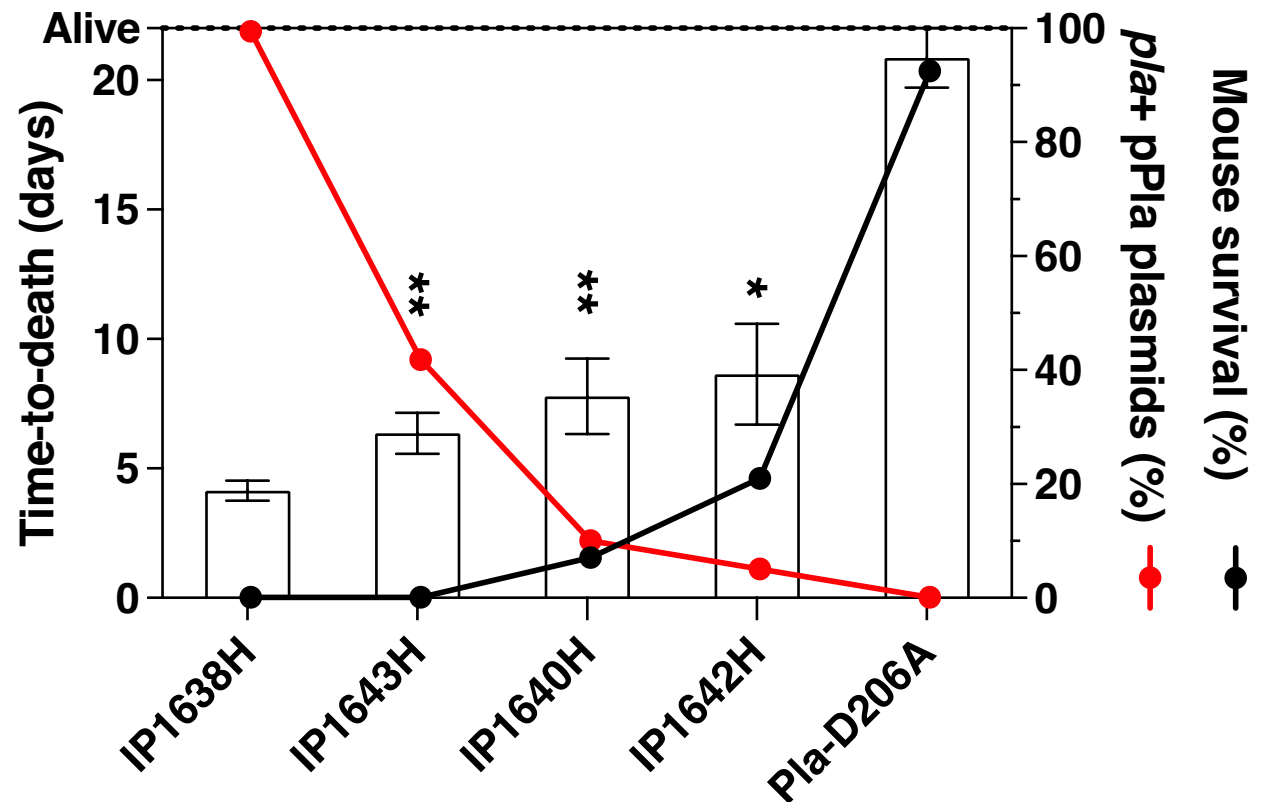


What the data show:

3. In mouse experiments the pla depletion leads to a strong attenuation of virulence and a massive drop in mortality

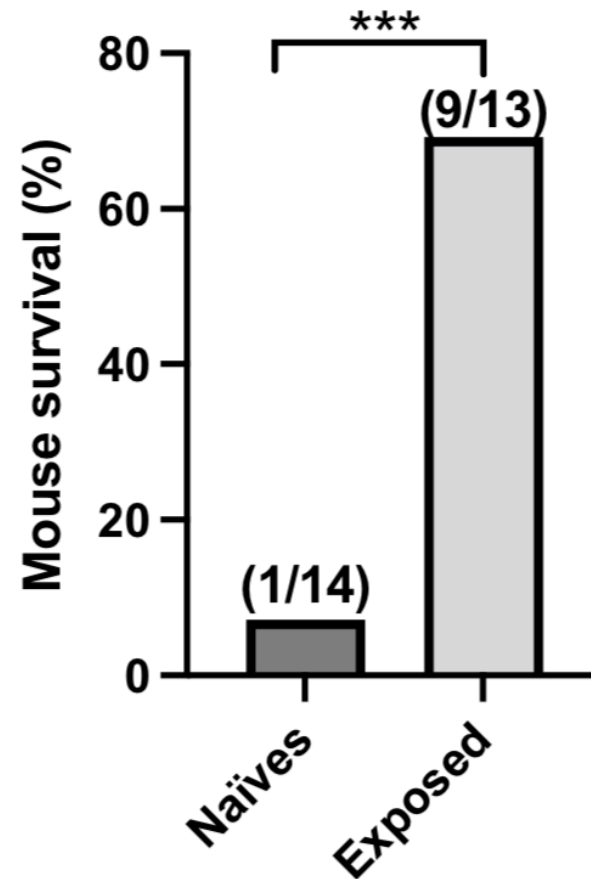
Lethality correlates with extent of *pla* depletion

- Mice subcutaneously infected with *pla* depleted strains survive longer than those infected with WT strains



Infection with *pla* depleted strains provides protection against plague

- Exposure to *pla* depleted strains leads to protection against subsequent infection



What the data show:

5. The fact that in each pandemic the starting strain is *always* pla + (wild type) suggests that the natural sylvatic reservoir could NOT have been rats or mice, but has to be something else.

What the data show:

6. The *Y. pestis* lineage from the second pandemic that ultimately led to the third pandemic was a *pl*⁺ (wild type), so it must have retreated to a non-rat reservoir.

What the data show:

6. The *Y. pestis* lineage from the second pandemic that ultimately led to the third pandemic was a *pla* + (wild type), so it must have retreated to a non-rat reservoir.
7. Depleted version of *pla*, plague strains survive for 3 centuries in the first and second pandemics.

How can we understand the “why?”

- What is the selective pressure which allows for this to happen?
 - Bubonic to Pneumonic plague?
 - Reservoir switch?
 - Human immune selection?

What can we infer from the data?

1. Rats, often associated as the driver of the Black Death and second pandemic of plague, may have been an amplification host at the onset due to exceptionally high densities living in close proximity to humans. (Side note - Rats are exceptional fecund animals, with on average 3 litters a year and between 6-22 offspring per litre. However they only live 12-16 months).

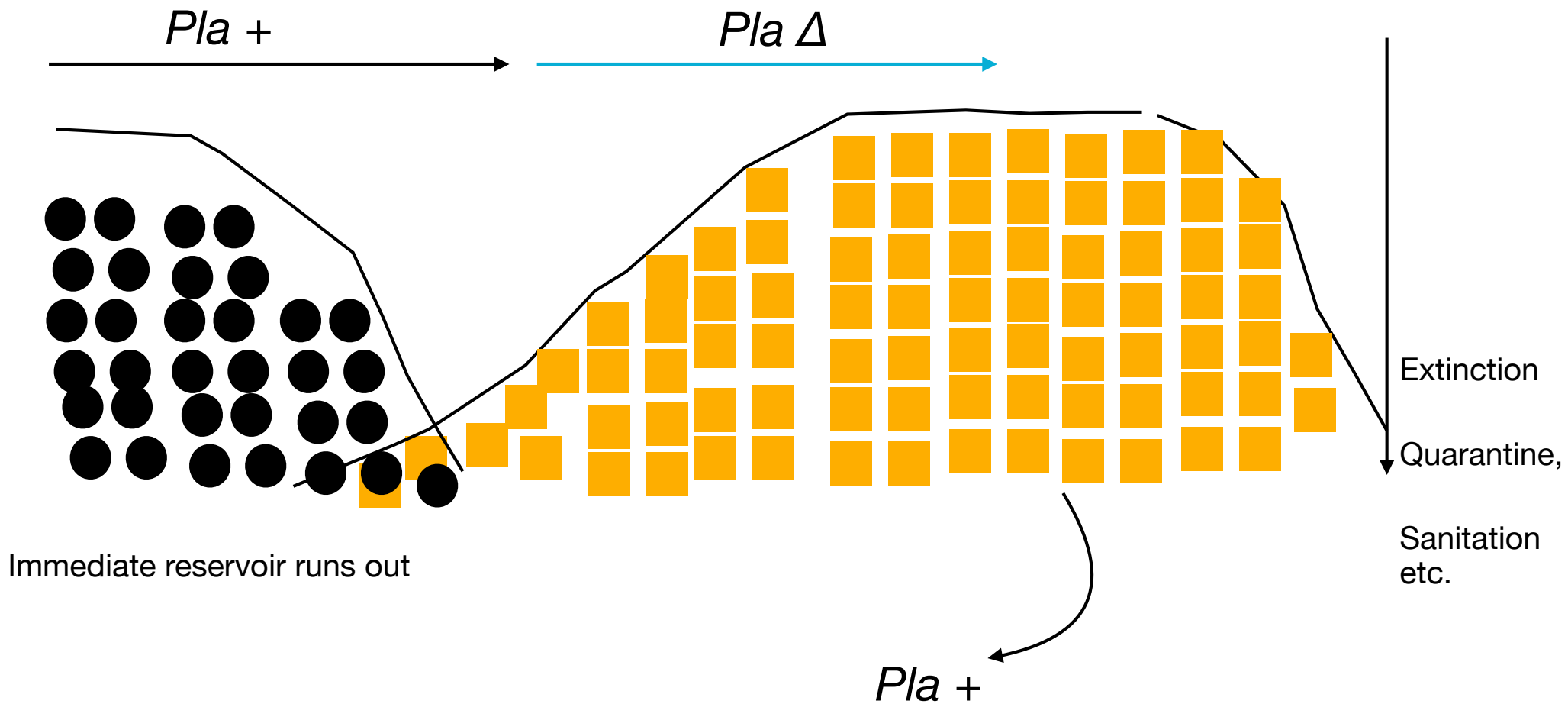
What can we infer from the data?

2. Despite our experiments which show that mice which survive an infection with an attenuated strain of Yp, are in turn 'resilient' to fully virulent plague, is a misdirect, as the lifespan is too short to establish a meaningful 'recalcitrant' population that 'reflects' plague.

Suggesting... the high rate of population turnover in rats suggest that it is population size/density that drives dynamics, not immunity]

What can we infer from the data?

3. Rat populations, under constant die offs, drop in density at some point into the second pandemic (and by assumption the first pandemic). We are suggesting that that die off coincides with the depletion of pla.



What can we infer from the data?

4. Rat populations have to be of sufficiently high density to survive under the pressure of high mortality, fully virulent plague. At extremely high densities where population replacement is easily regenerated, they are constantly supplying a new set of susceptible rodents that is of sufficient size/density for plague perpetuation and the pandemic cycling continues (i.e. rat populations are the drivers).

What can we infer from the data?

5. As the rat population crashed/declined, and number of susceptible hosts dropped, there was a strong selection for a reduction in virulence, thereby maintaining transmission. Rats, needed to live longer to be able to still maintain and perpetuate Yp transmission.

What can we infer from the data?

5. As the rat population crashed/declined, and number of susceptible hosts dropped, there was a strong selection for a reduction in virulence, thereby maintaining transmission. Rats, needed to live longer to be able to still maintain and perpetuate Yp transmission.

Or -Alternatively, the selection for reduced virulence could be for another commensal rodent, living a low density- and also highly susceptible to plague - house/door mouse (other?).

What can we infer from the data?

6. The 'extinction' of pla depleted strains at the end of the first and second pandemic has nothing to do with the depletion per se, but rather represents the collapse of the rodent reservoir, most likely be due to human behavioural changes, sanitation in cities, quarantine etc.

A Plasminogen-Activating Protease Specifically Controls the Development of Primary Pneumonic Plague

Wyndham W. Lathem,¹ Paul A. Price,¹ Virginia L. Miller,^{1,2} William E. Goldman^{1*}

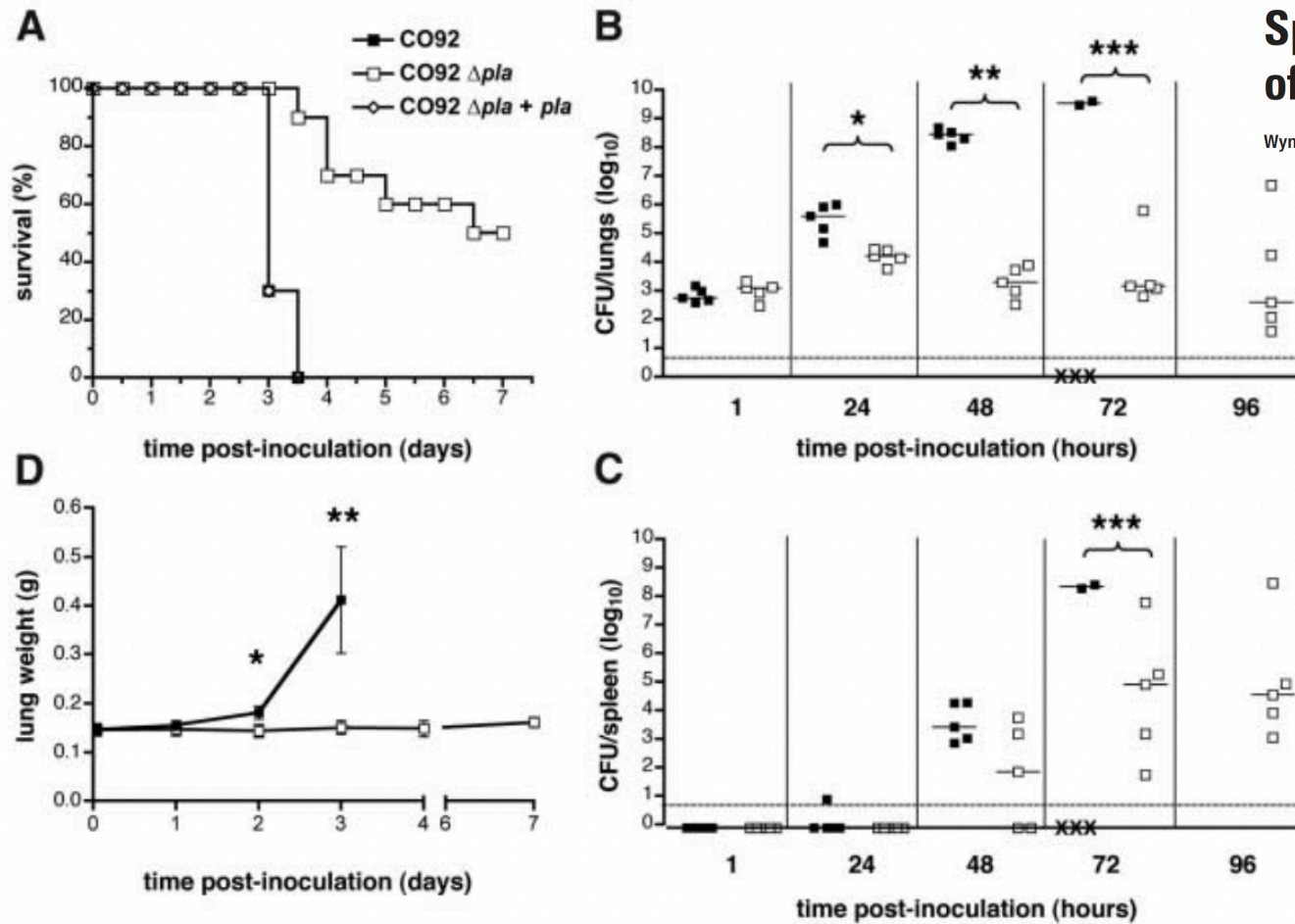
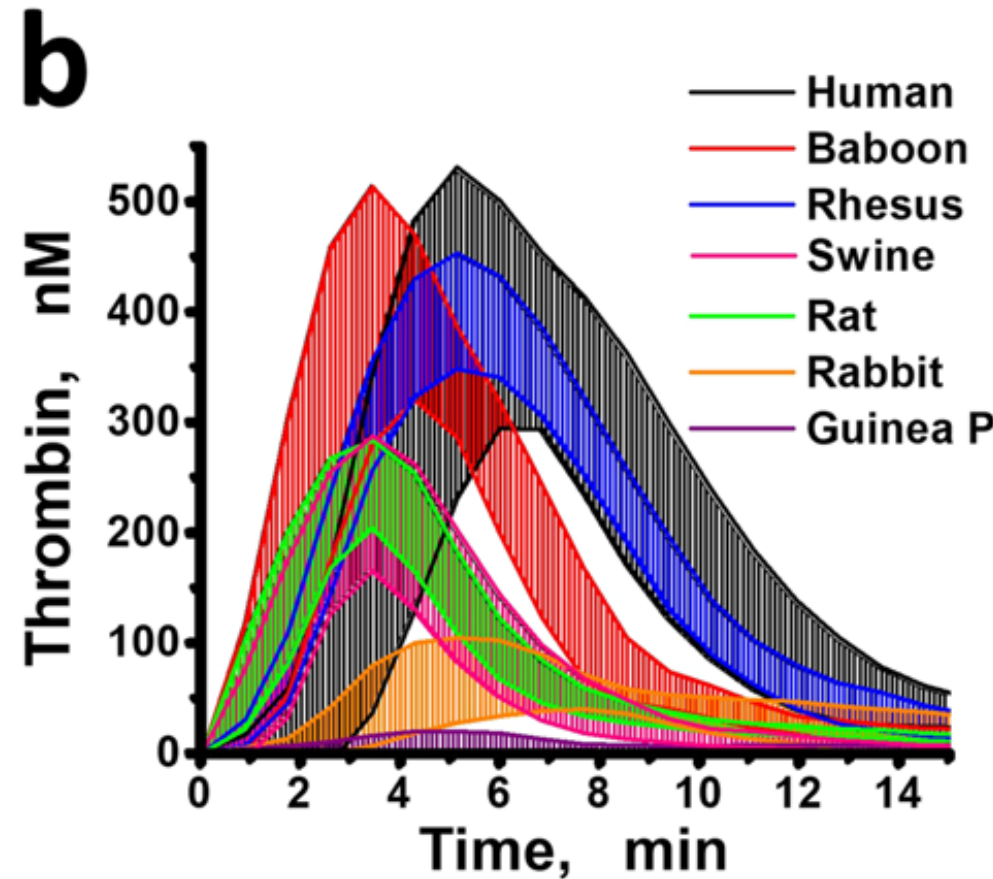
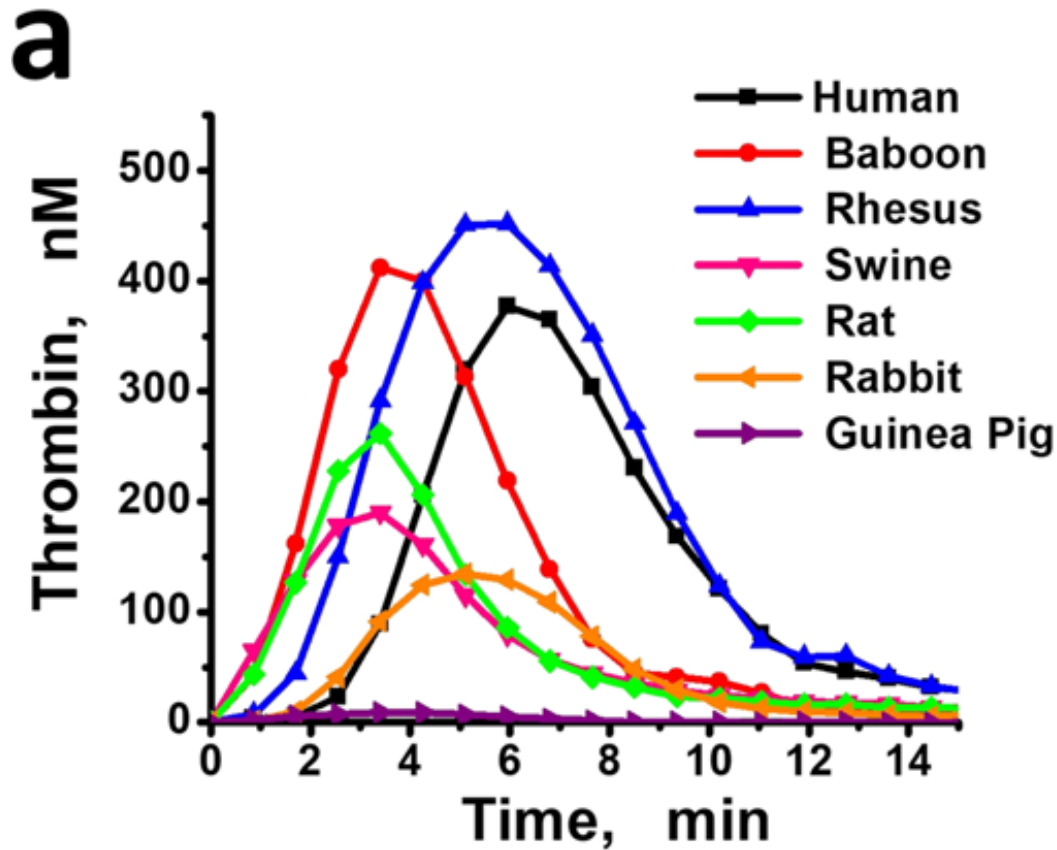


Fig. 1. Pla is required for *Y. pestis* to cause a fulminant infection of the lungs. **(A)** Survival of C57BL/6 mice infected intranasally with *Y. pestis* CO92 (black squares), CO92 Δpla (white squares), or CO92 Δpla complemented with *pla* (white diamonds). **(B and C)** Kinetics of infection with *Y. pestis* CO92 (black) or CO92 Δpla (white). Bacteria were introduced intranasally, and at various times CFU per organ in the lungs (B) and spleen (C) were determined. Each point represents the numbers recovered from a single mouse. The limit of detection is indicated by a dashed line. Symbols below the limit of detection represent mice that survived but did not have detectable numbers of bacteria; an "X" indicates a mouse that succumbed to the infection. A solid line indicates the median of CFU recovered. * $P = 0.037$, ** $P = 0.002$, and *** $P < 0.001$ (unpaired t test). **(D)** Gross weight of lungs from mice infected with *Y. pestis* CO92 (black) or CO92 Δpla (white). C57BL/6 mice were infected as above, and at various times lungs were excised and weighed. * $P = 0.01$ and ** $P < 0.001$. Each experiment was repeated twice. Error bars represent standard deviation.

Redoing these experiments with Black and Brown rats and other commensal rodents from Europe and the Middle East..



Can we model the rat population under these conditions?

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Correspondence and requests for materials should be addressed to L.D.H.
(e-mail: l.d.hurst@bath.ac.uk).

Metapopulation dynamics of bubonic plague

M. J. Keeling* & C. A. Gilligan†

* Department of Zoology, University of Cambridge, Downing Street, Cambridge CB2 3EJ, UK

† Department of Plant Sciences, University of Cambridge, Downing Street, Cambridge CB2 3EA, UK

Bubonic plague is widely regarded as a disease of mainly historical importance; however, with increasing reports of incidence^{1–3} and the discovery of antibiotic-resistant strains of the plague bacterium *Yersinia pestis*⁴, it is re-emerging as a significant health concern^{5,6}. Here we bypass the conventional human-disease models, and propose that bubonic plague is driven by the dynamics of the disease in the rat population. Using a stochastic, spatial metapopulation model, we show that bubonic plague can persist in relatively small rodent populations from which occasional human epidemics arise, without the need for external imports. This explains why historically the plague persisted despite long disease-free periods, and how the disease re-occurred in cities with tight quarantine control. In a contemporary setting, we show that human vaccination cannot eradicate the plague, and that culling of rats may prevent or exacerbate human epidemics, depending on the timing of the cull. The existence of plague reservoirs in wild rodent populations has important public-health implications for the transmission to urban rats and the subse-

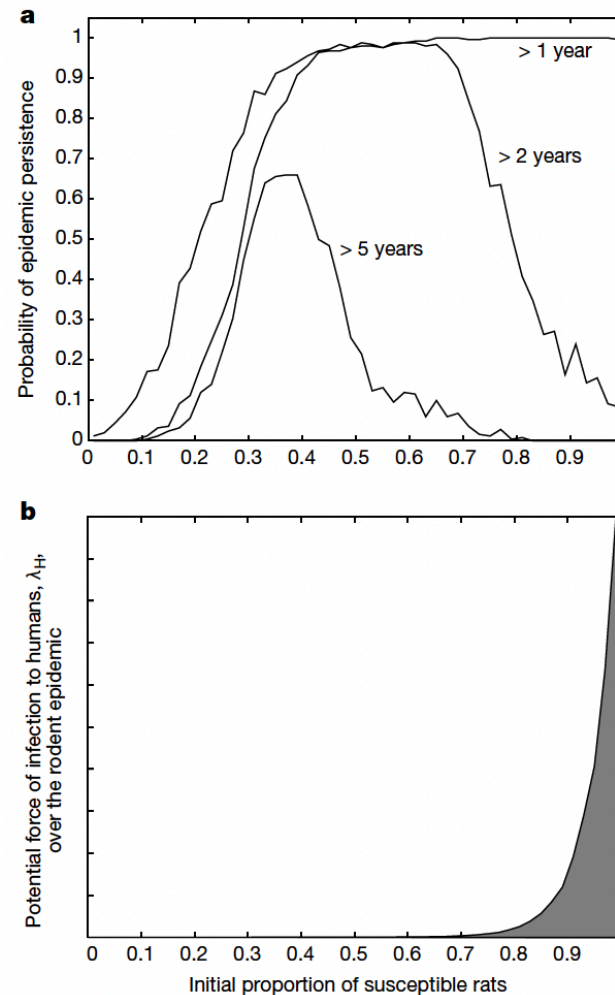


Figure 1 Results from 250 simulations of the stochastic epizootic model of bubonic plague with a rat population of 2,500. **a**, The probability that an infectious import generates an epidemic/endemic that lasts for more than 1, 2 or 5 years in the rat population. If the disease persists for more than 5 years it is likely to be in the endemic state. **b**, The potential force of infection for humans over the entire outbreak in rats ($\int_0^\infty \lambda_H(t) dt$), measured as the total number of infectious fleas that fail to find a suitable

letters to nature

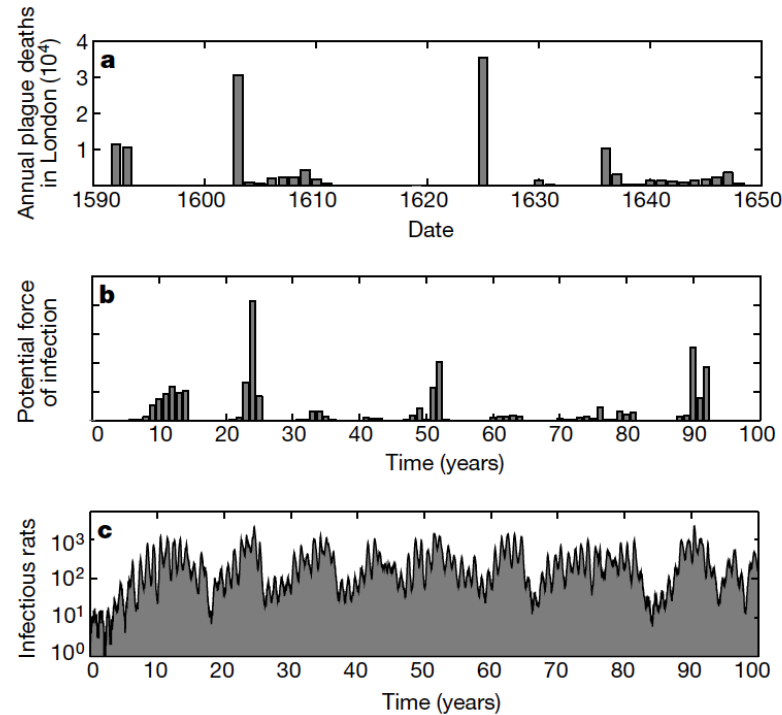


Figure 2 Comparison between historical data and the results from the metapopulation model. **a**, The annual deaths attributed to bubonic plague in London (1592–1661)²⁰. **b** and **c**, Results from a stochastic simulation of 100 years, with a rodent population of 60,000 distributed over 25 subpopulations (see Methods). **b**, Potential force of infection for humans showing erratic epidemics; for comparison with the historical data, yearly aggregates are shown. **c**, Number of infectious rats from the same simulation, showing how plague can persist in an isolated but spatially heterogeneous metapopulation.

forced to feed on alternative hosts such as humans, and a human epidemic occurs. (Although the pneumonic form of the disease can be passed directly from human to human, historical evidence suggests that this is rare^{8,16}, and is only triggered as a consequence of many rat-related cases.) This informal set of rules can be made

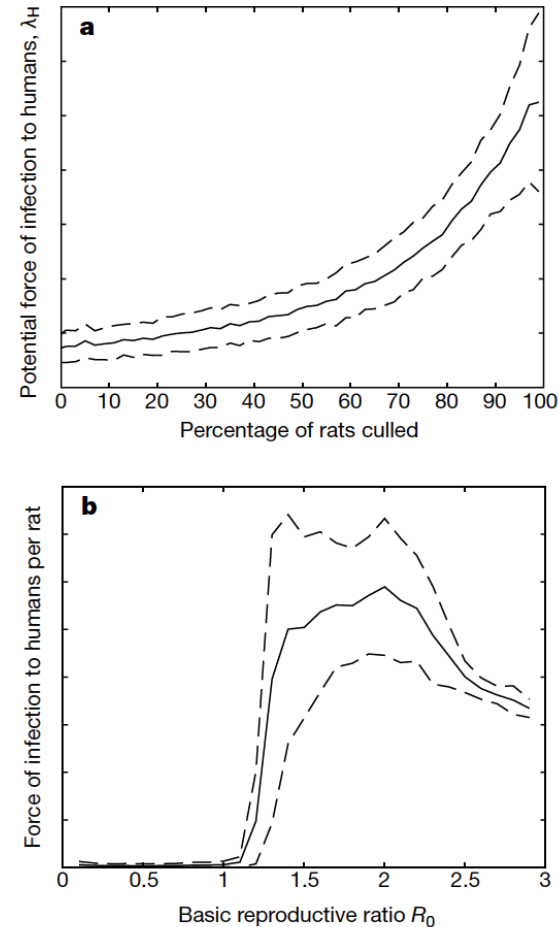


Figure 3 Results from the stochastic model for a single rat subpopulation. Solid lines, means; dashed lines, 95% confidence intervals. **a**, The effect of culling the rats once human cases have been detected. **b**, The potential force of infection to humans per rat over the entire outbreak shows clear threshold behaviour. The basic reproductive ratio R_0 is defined as $R_0 = \beta_R K_F / d_F [1 - \exp(-aK_R)]$. Different values of R_0 were obtained by varying the flea-searching efficiency, **a**, and the rat carrying capacity, K_R .